

Does Model Complexity Pay Off?

A Rolling-Origin Benchmark for Forecasting German Household Overnight Deposit Volumes

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July 30, 2026

Abstract

Liquidity management ensures that a bank has sufficient cash to fulfill its short-term obligations while minimizing the cost of holding excess liquidity, and non-maturity deposits (NMDs) are among the most important sources of this liquidity. Despite their economic importance, forecasting NMD volumes remains methodologically underdeveloped, with no established consensus on the optimal modeling approach. This study compares two naive benchmarks, a regularized linear model, gradient-boosted trees, a feedforward neural network, and a Temporal Fusion Transformer for forecasting German household overnight deposit volumes over a 100-day horizon, using a rolling-origin evaluation across four forecast origins between 2009 and 2025 that represent distinct interest-rate regimes. All models share an identical information set with a 100-day lookback window, an identical tuning budget, and lag-augmented covariates. The regularized linear model achieved the lowest mean forecast error ($\text{MAE} = 9.36 \text{ Bn. €}$), the lowest error relative to the constant-level benchmark (0.44), and was the only model with a positive mean R^2 (0.178), while the feedforward network and gradient-boosted trees did not improve on a one-step drift benchmark. Because only four origins are available, these differences are reported descriptively rather than as significance tests. The results indicate that increased model complexity does not translate into higher forecasting accuracy for this task, with direct implications for how banks select forecasting tools for liquidity management.

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1 Introduction

The Basel Committee on Banking Supervision (?) defines non-maturity deposits (NMDs) as liabilities without a contractual maturity where the depositor is free to withdraw at any time. They consist mainly of fully transferable deposits and non-transferable deposits that can be converted by end of the next business day. The interest paid on NMDs is significantly below other products issued by banks. Moreover, since there is no fixed maturity date, banks reserve the right to change this interest rate at any time. Therefore, because of their stability and low cost, NMDs are an important source of funding for banks. Especially for German banks, NMDs nowadays represent the most important funding source after exhibiting dynamic growth in the aftermath of the financial crisis of 2008 (?). Considering this, it becomes clear how important the study of NMDs is for bank liquidity management, the process of assessing the liquidity needs of a bank and supplying or absorbing the appropriate amount of liquidity through open market operations (?). However, because of their stochastic nature, the study of these products is extremely challenging. Despite its importance, academic research on this topic is still scarce. Recent years have seen growing interest in the subject in the form of graduate theses, journals and papers (??). Nevertheless, the current literature presents a wide range of approaches, from economic to statistical and mathematical models, and there is yet to be a generally accepted scientific consensus on the optimal methodology for modeling NMDs. Hence, the need for a structured and effective methodology for forecasting these financial products is evident.

This study addresses the identified research gap by developing a comparative framework for forecasting non-maturity deposit volumes. We evaluate naive benchmarks, statistical models, and machine-learning models across alternative covariate sets and multiple forecast origins representing different economic environments. Although the study is motivated by NMD modelling, the empirical analysis uses publicly available German household overnight deposit data as a proxy for aggregate household NMD volumes; Section 6.6 discusses what this aggregation does and does not permit us to conclude.

The study makes three contributions. First, it examines whether greater model complexity leads to more accurate forecasts under a protocol in which all models receive an identical information set and an identical tuning budget. Second, it evaluates how interest-rate, macroeconomic, and financial-market covariates affect forecasting performance. Third, it investigates whether model performance remains stable across different economic periods, using both absolute and scale-free error measures, since the deposit stock itself grows strongly over the sample.

The remainder of this paper is structured as follows. Section 2 reviews the literature on NMD modeling and on the more general question of whether complex forecasting models outperform simple ones, and positions this study within that research. Section 3 presents the data and preprocessing methodology, including variable definitions, interpolation, and target transformation. Section 4 describes the forecasting framework, the models under comparison, the tuning protocol, and the rolling-origin evaluation design. Section 5 reports and compares the forecasting results. Section 6 discusses these findings, their limitations, and their managerial implications. Finally, Section 7 concludes and outlines directions for future research.

Research Question(s):

RQ1 How accurately can statistical and machine-learning models forecast German household overnight deposit volumes over a 100-day horizon, measured both in absolute terms and relative to a naive benchmark?

RQ2 Do nonlinear machine-learning models outperform linear and naive benchmark models under the same rolling-origin evaluation framework and the same tuning budget?

RQ3 Which groups of covariates provide the most improvement in forecasting performance, and is that improvement stable across economic regimes?

2 Related Work

Modelling non-maturity deposits. The academic treatment of NMDs originates in the valuation literature rather than in forecasting. ? develop an arbitrage-free term structure framework in which deposit balances and deposit rates are modelled jointly as functions of the market rate, allowing the deposit portfolio to be priced as a contingent claim. ? model deposit balances under imperfect competition and show that the sensitivity of volumes to market rates depends on the bank’s market power rather than on the rate level alone. ? estimates a partial-adjustment specification in which balances converge only slowly to their long-run equilibrium, which motivates the strong autocorrelation that any short-horizon forecasting model must handle. A second strand translates these dynamics into replicating-portfolio techniques for asset–liability management (????): rather than forecasting the deposit path directly, a static or dynamic portfolio of fixed-income instruments is constructed whose cash flows mimic the deposit book. This is the approach embedded in current supervisory practice (??), where retail NMDs are subject to caps on the assumed average repricing maturity. Common to all of these contributions is that the volume process is a modelling *input* whose out-of-sample predictability is rarely evaluated in its own right.

Machine learning for NMD volumes. The most direct predecessor of this study is ?, who compares a Lévy-driven Ornstein–Uhlenbeck process with an LSTM network on German deposit data and provides the interpolation and €STR reconstruction procedures that we adopt in Section 3. ? likewise apply machine-learning methods to NMD modelling. Both studies evaluate on a single, fixed out-of-sample window. This matters, because a single window cannot separate a genuinely superior model from a model that happens to suit one economic environment (?).

Does complexity pay off? The broader forecasting literature provides a strong prior against complexity. The M4 and M5 competitions (??) found that pure machine-learning submissions were systematically outperformed by statistical and hybrid methods, and that the winning entries combined simple components. ? show that a single linear layer matches or beats several published Transformer architectures on standard long-horizon benchmarks, and argue that self-attention destroys the temporal ordering that these tasks depend on. ? document that many reported neural-network gains in time-series forecasting do not survive a rigorous rolling-origin protocol with matched tuning effort. Our results are consistent with this body of work and extend it to a financial-liability series that has not, to our best knowledge, been examined under such a protocol.

Positioning. Table 1 summarises how this study differs from the most closely related work. We are not aware of a study that benchmarks naive, regularized linear, tree-based, and attention-based models on NMD volumes under a common rolling-origin design, a common information set, an identical tuning budget, and an explicit scale-free error measure.

3 Data and Preprocessing

3.1 Target Variable and Explanatory Variables

Detailed non-maturity deposit (NMD) data are generally not publicly available because they contain sensitive bank information. For this study, we use the publicly available end-of-period volume of overnight deposits held by private households, published by the Deutsche Bundesbank (?), as a proxy for aggregate NMD volumes. Household overnight deposits provide an accurate publicly observable proxy since they are a subset of NMDs and account for a substantial share of banks’ total deposit funding (?). Therefore, the aggregate volume of household overnight

Table 1: Positioning of this paper relative to related work on NMD modelling and on model complexity in forecasting.

Reference	Modeling approach	Multiple origins	Matched tuning
?	Arbitrage-free valuation of deposits	✗	—
?	Balance dynamics under imperfect competition	✗	—
?	Partial-adjustment balance model	✗	—
?	Replicating portfolios	✗	—
?	Deposit pricing and volume dynamics	✗	—
?	Machine learning for NMD modelling	✗	✗
?	Lévy-driven OU process vs. LSTM	✗	✗
?	Large-scale competition, generic series	✓	✓
?	Linear layer vs. Transformers	✓	✓
This paper	Naive, Ridge, XGBoost, FFN, TFT on NMD volumes	✓	✓

deposits is likely to respond to several of the same economic drivers. The dataset covers the period January 2003 to September 2025 and the series is measured in billions of euros.

Euro Short-Term Rate (€STR): sourced from the ECB (?), is available at daily frequency and captures the average overnight borrowing rate of banks in the euro area. The reasoning for including €STR as a covariate is that a shift in it could cause a shift in the rates banks apply to deposit accounts, meaning a change in total overnight deposits.

German geopolitical risk index (GPRC_DEU): published on a monthly basis (?), quantifies geopolitical risk in Germany using automated text search in the U.S. newspapers *New York Times*, *Chicago Tribune* and *Washington Post*. The geopolitical context may have an influence because as uncertainty rises, customers might be more inclined to postpone spending and leave cash in their accounts, leading to an increase in overnight deposits.

Month-on-month inflation (MoM-Inflation): sourced from Destatis (?), calculates the percentage change in the Consumer Price Index relative to the preceding month. Inflation changes how much households spend and save, and therefore may help explain movements in their total overnight deposits.

DAX daily closing prices: obtained via the yfinance Python library (?), retrieving data from Yahoo Finance (?) for the ticker ^GDAXI. As the DAX reflects changes in the German stock market as well as investor confidence, this may influence the total amount held by private households.

10-year German government bond yield (10Y Bond): sourced at business-day frequency from the Deutsche Bundesbank (?). As the ten-year government bond is a relatively safe investment product, private households may be drawn to invest more in it if the rate is sufficiently attractive, investing less in overnight deposit accounts.

Deposit Rate: refers to the household deposit interest rate, sourced from the Deutsche Bundesbank (?). The more interest households receive on deposits, the more attractive it may be to keep their money in overnight deposit accounts rather than seeking alternative investments.

3.2 Data Alignment and Transformation

Since multiple variables are published at monthly frequency, they require interpolation to a daily grid before model training. Deposit volume, the deposit rate, inflation, and geopolitical risk are converted from their monthly series to daily frequency following piecewise cubic Hermite interpolating polynomial (PCHIP) interpolation (?, Ch. 5.2). This preserves local monotonicity of the input series and avoids artificial overshoots between monthly observations (?). We stress that this transformation is a modelling convenience and not a neutral operation on the target: the resulting daily target is a smoothed construct whose effective information content remains

monthly. Its consequences for the interpretation of all reported metrics are discussed explicitly in Section 6.6.

As each published €STR value reflects the previous business day’s transactions, it constitutes a direct, transaction-based measure of interbank market conditions. While €STR was officially introduced in October 2019, the sample period extends back to January 2003. For the period March 2017 to September 2019, the Pre-€STR (?) is used as a direct predecessor series. For earlier observations, the series is reconstructed following ?, Ch. 5.2.

For variables available at business-day frequency (€STR, DAX and 10Y Bond), weekend and public holiday gaps are handled via forward-fill, carrying the last observed value forward to the following business day. This ensures a complete daily index across the full sample period while avoiding the introduction of artificial variation on non-trading days.

Figure 1 illustrates all variables over the full sample period after interpolation to daily frequency.

3.3 Lagged Covariates

Economic covariates do not act on deposit behaviour instantaneously. A change in the policy rate is passed through to retail deposit rates with a delay, and households reallocate balances only gradually once that pass-through has occurred; the partial-adjustment evidence of ? points in the same direction. A specification that contains only the contemporaneous value $\mathbf{x}_t$ implicitly assumes an immediate effect and therefore cannot represent this mechanism, even if it is present in the data.

We therefore augment every covariate set with lagged copies of each covariate at $\ell \in \{1, 5, 20\}$ trading days, corresponding to a one-day, one-week, and roughly one-month delay. For a covariate set $\mathcal{C}$, the feature vector at time t is

$$\mathbf{x}_t = (\{c_t\}_{c \in \mathcal{C}}, \{c_{t-1}\}_{c \in \mathcal{C}}, \{c_{t-5}\}_{c \in \mathcal{C}}, \{c_{t-20}\}_{c \in \mathcal{C}}). \quad (1)$$

The contemporaneous value is retained alongside the lags so that immediate and delayed effects can both be represented and the model, rather than the analyst, selects between them. Shifting is performed positionally on the trading-day index, so a lag of five denotes five trading days rather than five calendar days; the first $\max_{\ell} \ell = 20$ rows of the sample are lost.

Because the lookback window itself spans 100 trading days (Section 4.1), the explicit lag features are strictly nested within the information the models already receive. Their purpose is not to extend the reach of the information set but to make delayed relationships directly representable: a flattened window presents the same value at 100 distinct positions, and a lag feature states explicitly which displacement is economically meaningful, which is a substantially easier learning problem for models without an internal notion of temporal displacement. Lag augmentation therefore applies to Ridge, XGBoost and the FFN; the TFT processes the covariate history sequentially through its encoder and does not require explicit lag columns.

3.4 Target Transformation

As evident from Figure 1, the deposit-volume level exhibits a clear upward trend throughout the timeline, indicating nonstationarity in the mean. First differencing is therefore applied to obtain a stationary target series and to limit accumulating directional errors (?), as illustrated in Figure 2. Second differencing was rejected due to over-differencing.

After interpolation to daily frequency, the dataset comprises 8,276 daily observations covering January 2003 to September 2025. Because the target is published monthly, these daily rows correspond to approximately 273 independent monthly releases. The effective sample size is therefore roughly thirty times smaller than the nominal number of rows, a point we return to in Section 6.6.

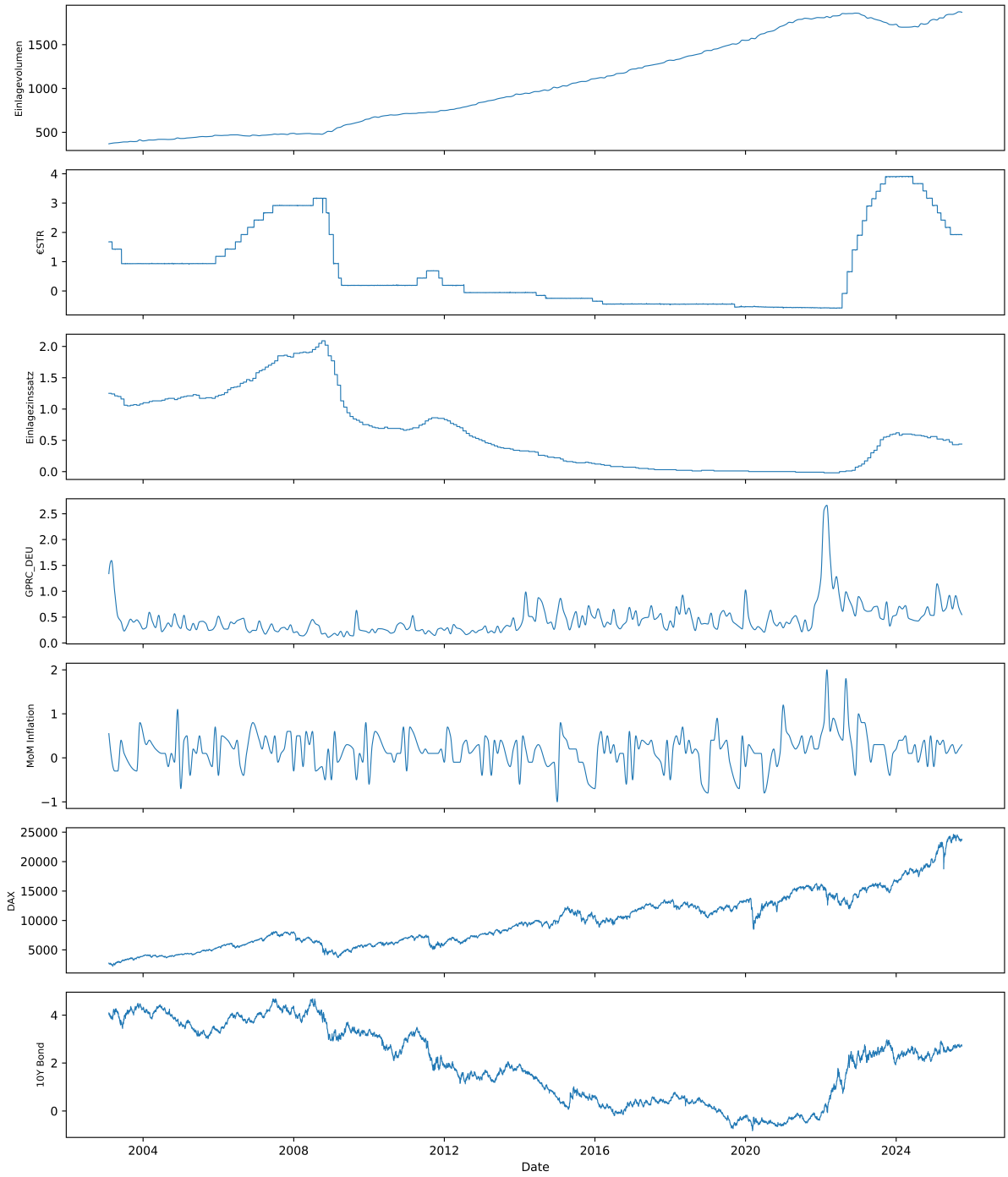


Figure 1: Time series of all variables over the sample period January 2003 – September 2025. Vertical axes are in the native unit of each series (deposit volume and DAX in levels, all rates in percentage points, the geopolitical risk index in index points). All monthly series are shown after PCHIP interpolation to daily frequency. The €STR series is observed from October 2019 onward and reconstructed backwards for earlier periods as described in the text.

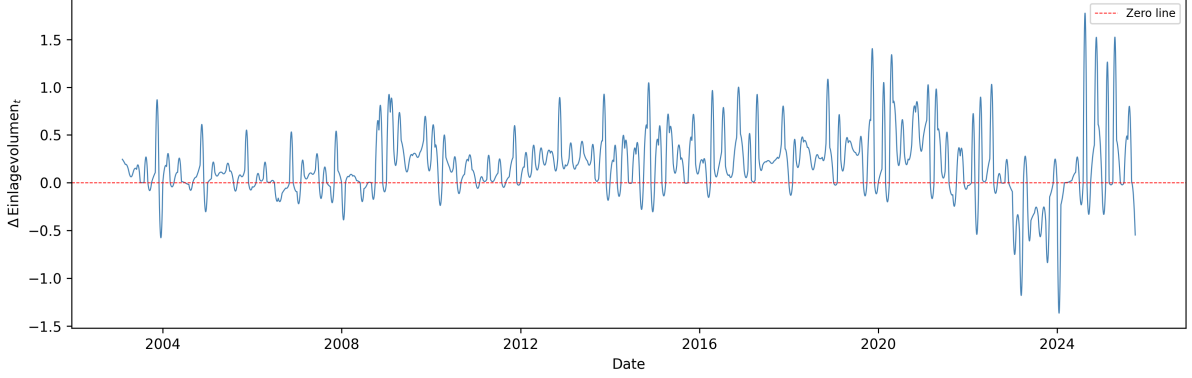


Figure 2: First difference of the deposit volume, Δy_t , in EUR billions per day (vertical axis) over the full sample period January 2003 – September 2025 (horizontal axis). The series fluctuates around zero throughout, which is consistent with stationarity after differencing. Increased volatility from 2022 onward coincides with the ECB interest-rate tightening cycle.

4 Forecasting Models

4.1 Forecasting Framework

Let y_t denote the volume of German household overnight deposits at time t , and let $\mathbf{x}_t$ denote the lag-augmented covariate vector of Equation (1). The objective is to forecast deposit volumes over a 100-day horizon using the information available during the preceding 100 days. The horizon $H = 100$ corresponds to roughly one quarter of trading days and thus to the planning cycle used in operational liquidity management.

At the time of forecast t , the models use a lookback window of $L = 100$. The information set therefore contains the historical deposit volume and the selected covariates over the period $t - L + 1, \dots, t$. Each model produces the complete vector of predicted changes for the following $H = 100$ days:

$$\widehat{\Delta \mathbf{y}}_{t+1:t+H} = f(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}), \quad L = 100, \quad H = 100. \quad (2)$$

Setting $L = H$ follows the common convention that the context available to a multi-horizon forecaster should be at least as long as the horizon it must predict (??). In our setting the choice is also dictated by the effective resolution of the target: because the deposit series is published monthly and interpolated to a daily grid (Section 3.2), a window of 100 trading days spans approximately five genuine observations of the target, whereas a shorter window would span only one or two and would leave the models with essentially no information about the shape of the recent trajectory.

The choice is not costless, and we state the cost explicitly because it interacts with our central claim. Each training example consumes $L + H = 200$ consecutive observations, so the number of usable training sequences at origin o_k is $o_k - L - H + 2$. At the January 2009 origin, which offers roughly 1,550 trading days of history, this leaves approximately 1,350 overlapping sequences that are themselves derived from only about sixty independent monthly releases. Parameter-rich models are therefore fitted on a small effective sample precisely at the origin that the results identify as the hardest (Section 5), which offers an additional explanation for the poor performance of the flexible architectures there beyond the economic argument of Section 6.4.

Finally, even a 100-day window does not span an annual cycle of approximately 250 trading days, so no model in this comparison can learn calendar seasonality or multi-year interest-rate cycles from its encoder input. This restriction applies identically to all models and therefore does not bias the comparison between them, but it does bound what any of them could in principle achieve; Section 6.6 treats it as a threat to the external validity of our conclusion.

The study applies a direct multi-output forecasting strategy: each model estimates all 100 future changes simultaneously rather than recursively one day at a time. The predicted changes are then converted back into deposit-volume levels according to

$$\hat{y}_{t+h} = y_t + \sum_{j=1}^h \widehat{\Delta y}_{t+j}, \quad h = 1, \dots, H. \quad (3)$$

Only information available up to time t is used for model estimation and for fitting the feature scaler, which is a robust scaler estimated on the training window alone. Actual covariate values within the 100 forecast days are never used. For the TFT, covariates are held constant at their most recently observed value over the horizon, while date-based variables that are known in advance are supplied for the entire forecast horizon.

4.2 Model Specifications

Naive benchmarks. Two naive benchmarks are used, and we name them by their construction rather than by the ambiguous label “random walk”. The *one-step drift* benchmark extrapolates the single most recently observed daily change at a constant rate,

$$\hat{y}_{t_0+h} = y_{t_0} + h \cdot \Delta y_{t_0}, \quad h = 1, \dots, 100,$$

that is, an ARIMA(0, 1, 0) process whose drift is estimated from one observation rather than from the training average. The *constant-level* benchmark, which is the random walk without drift in the conventional sense, assumes the level remains unchanged, $\hat{y}_{t_0+h} = y_{t_0}$. Both benchmarks are fully interpretable, require no covariates, and serve as the reference against which the estimated models are compared. Because the interpolated target is locally very smooth (Section 3.2), the one-step drift benchmark is a demanding reference at short horizons and a fragile one at long horizons, which is visible in the results.

Linear regression (Ridge). A multi-output ridge regression maps the flattened 100-day lookback window to the 100-step forecast vector, regularized by an ℓ_2 penalty α selected via time-series cross-validation. As a linear model, Ridge yields directly inspectable coefficients. The flattened design matrix is high-dimensional – 100 time steps times the number of covariates and their lags – and heavily collinear, which makes shrinkage not merely convenient but structurally necessary.

XGBoost. Gradient-boosted regression trees are trained on the same flattened input representation (?), with maximum depth, learning rate and subsample ratio selected by chronological time-series cross-validation. Boosted trees can capture non-linear covariate interactions that Ridge cannot (?, Ch. 10), while remaining moderately interpretable.

Feedforward neural network (FFN). The FFN is a multilayer perceptron with ReLU activations trained to predict the 100-step output vector directly. Contrary to a common shortcut in benchmark studies, the FFN is *tuned under the same protocol and the same budget* as Ridge and XGBoost (Section 4.3); its layer configuration and ℓ_2 penalty are selected by the same chronological cross-validation rather than fixed a priori. Training uses the Adam optimizer with early stopping on the chronologically last 15% of the training window, with a patience of 20 epochs and a cap of 500 epochs. Using a chronological rather than a shuffled validation split matters here, because with $L = 100$ the training sequences overlap heavily: a shuffled split would place later windows in the training set and earlier, overlapping windows in validation, leaking future information into the early-stopping decision.

Temporal Fusion Transformer (TFT). Unlike the models above, the TFT explicitly distinguishes between *known* future inputs and *unknown* future inputs (?). The known future inputs are calendar features that are exactly determined for every future date – weekday, month-end indicator, and day of month – and are computed from the actual future dates rather than carried forward. The unknown future inputs are the macro-financial covariates, which are held at their last observed value over the horizon. The encoder length is set to $L = 100$ to match the other models, and the model is trained with early stopping on validation loss with a patience of five validation checks. Despite being the most complex model considered, its attention mechanism was explicitly designed to offer a degree of interpretability that purely black-box models lack.

Each model is fitted $S = 5$ times to form an ensemble. Ridge, FFN and XGBoost use bootstrap-resampled training sequences together with different random seeds; repeated TFT fits differ in the random seed only. Ensemble members are combined by averaging their forecasts.

4.3 Tuning Protocol

A comparison of “complex” against “simple” models is only meaningful if the models receive comparable optimisation effort. We therefore impose a single tuning budget of at most twelve grid configurations per model, searched by grid search with a three-fold chronological `TimeSeriesSplit` and mean absolute error as the selection criterion. Hyperparameters are re-tuned at every forecast origin using only data available at that origin, so that no configuration selected on a later window is transferred to an earlier one. Table 2 documents the complete search space; the TFT architecture is fixed a priori at the configuration shown, since a grid search over the transformer was not computationally feasible within the same budget. This last asymmetry works *against* our headline conclusion rather than for it and is noted as a limitation in Section 6.6.

Table 2: Hyperparameter search spaces and fixed settings. All searched models stay within the common budget of twelve configurations.

Model	Search space / fixed settings	Configurations
Ridge	$\alpha \in \{10^{-3}, \dots, 10^3\}$, 8 values log-spaced	8
FFN	hidden layers $\in \{(64, 32), (128, 64)\}$; ℓ_2 penalty $\in \{10^{-4}, 10^{-3}\}$; Adam, learning rate 10^{-3} ; early stopping, patience 20 epochs, max. 500	4
XGBoost	max. depth $\in \{3, 4, 5\}$; learning rate $\in \{0.05, 0.1\}$; subsample $\in \{0.7, 0.9\}$; 50 trees during search, 100 at final fit	12
TFT	fixed: hidden size 32, 4 attention heads, dropout 0.1, continuous hidden size 16, Adam, learning rate 10^{-3} , batch size 64, max. 40 epochs, early stopping patience 5, MAE loss	—
<i>Shared:</i> $L = 100$, $H = 100$, $S = 5$ ensemble members, robust scaling fitted on training window, seed 42.		

4.4 Covariate Specifications

To assess the predictive value of different economic information, each non-naive model is evaluated using the nine predefined covariate sets shown in Table 3. These specifications compare economically related variable groups, individual interest-rate variables, and subsets of the complete dataset. Each set is lag-augmented according to Equation (1). The final column reports the identifier used in the accompanying code and in the figure filenames, so that every figure in this paper can be traced to the specification that produced it.

4.5 Rolling-Origin Evaluation

Forecasting performance is assessed using an expanding-window rolling-origin design. At each forecast origin o_k , all observations available up to that date form the training sample, while the

Table 3: Covariate specifications and corresponding code identifiers.

Specification	Included variables	Code label
All	€STR, deposit rate, geopolitical risk, inflation, DAX, and ten-year German government bond yield	alle
Interest rates	€STR and deposit rate	nur_zinsen
Financial markets	DAX and ten-year government bond yield	nur_maerkte
Macroeconomic	Geopolitical risk and inflation	nur_makro
Rates and macroeconomy	€STR, deposit rate, geopolitical risk, and inflation	zinsen_makro
Excluding DAX	All variables except DAX	ohne_dax
Excluding inflation	All variables except inflation	ohne_inflation
€STR only	€STR	nur_estr
Deposit rate only	Deposit rate	nur_einlagezins

following $H = 100$ days form the evaluation window:

$$\mathcal{T}_k = \{1, \dots, o_k\}, \quad \mathcal{E}_k = \{o_k + 1, \dots, o_k + H\}.$$

The models are re-estimated, and their hyperparameters re-selected, at every origin as newly observed data becomes available. The evaluation uses four origins: 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025. These dates were selected to represent distinct interest-rate and macroeconomic regimes – the 2008/2009 financial crisis, the subsequent low-interest-rate period, the pre-COVID environment, and the aftermath of the post-2022 tightening cycle – rather than a dense monthly rolling scheme, in order to test model robustness across qualitatively different economic conditions. The earliest origin is additionally constrained by the training requirement of $L + H = 200$ observations per sequence discussed in Section 4.1. The price of this design choice is a very small number of evaluation points, which we address directly in Sections 4.6 and 5.5.

4.6 Evaluation Metrics

Performance is evaluated for both the predicted first differences and the reconstructed deposit-volume levels. Let $y_{k,h}$ and $\hat{y}_{k,h}$ denote the observed and predicted volume at horizon h after origin o_k . The metrics are

$$\begin{aligned} \text{MAE}_k &= \frac{1}{H} \sum_{h=1}^H |y_{k,h} - \hat{y}_{k,h}|, \\ \text{RMSE}_k &= \sqrt{\frac{1}{H} \sum_{h=1}^H (y_{k,h} - \hat{y}_{k,h})^2}, \\ R_k^2 &= 1 - \frac{\sum_{h=1}^H (y_{k,h} - \hat{y}_{k,h})^2}{\sum_{h=1}^H (y_{k,h} - \bar{y}_k)^2}, \end{aligned}$$

where $\bar{y}_k$ is the mean observed volume in evaluation window k . Negative R^2 values indicate performance below the test-window mean benchmark.

A scale-free measure. Absolute errors in EUR billions are not comparable across our four origins, because the deposit stock itself grows strongly over the sample: an error of 10 Bn. € in 2009 represents a far larger relative deviation than the same error in 2025. Averaging absolute MAE across origins therefore implicitly overweights the later, higher-level windows, and the resulting standard deviation conflates genuine model instability with a pure level effect. Following

the logic of scaled error measures (?), we additionally report the error relative to the constant-level benchmark at the same origin,

$$\text{relMAE}_k = \frac{\text{MAE}_k^{\text{model}}}{\text{MAE}_k^{\text{constant-level}}}, \quad (4)$$

so that $\text{relMAE} = 1$ denotes parity with a forecaster who assumes nothing changes, and values below one denote improvement. This quantity is dimensionless, comparable across regimes, and is the measure on which we base our answer to RQ1.

Reporting. Metrics are calculated separately at each forecast origin and summarized by their mean and standard deviation across origins; the complete per-origin matrix is reproduced in Appendix A rather than only aggregated, since with four observations an aggregate conceals more than it reveals. The primary comparison uses MAE on the reconstructed levels and relMAE; RMSE and R^2 provide complementary information, and metrics on first differences serve as diagnostics. A Diebold–Mariano test (?) with the small-sample correction of ? was implemented but is deliberately not applied to the main results: with four origins the test has essentially no power, and reporting a non-rejection would invite the false inference that the models are equivalent. Model comparisons are therefore reported descriptively throughout.

5 Results

5.1 Forecasting Performance

Table 4 reports, for each model, the covariate specification with the lowest mean level MAE across the four origins, together with the scale-free measure of Equation (4). The selection of the covariate set is discussed critically in Section 5.5.

Table 4: Forecasting performance across the four rolling origins (2009, 2013, 2019, 2025). MAE and RMSE are measured in EUR billions and reported as mean $\pm$ standard deviation across origins. relMAE is the mean ratio to the constant-level benchmark at the same origin (Equation (4)); values below one indicate improvement over that benchmark. Estimated models are shown with their best covariate specification; naive benchmarks use no covariates.

Model	Best covariate set	MAE	RMSE	Mean R^2	relMAE
<i>Estimated models</i>					
Ridge	All covariates	9.36 ± 11.11	10.26 ± 12.08	0.178	0.44 ± 0.31
TFT	Macroeconomic	10.78 ± 12.61	12.71 ± 14.18	−0.361	0.49 ± 0.39
FFN	Macroeconomic	11.85 ± 6.90	15.22 ± 10.35	−0.605	0.71 ± 0.34
XGBoost	Excluding inflation	12.71 ± 13.28	14.90 ± 14.79	−0.784	0.69 ± 0.64
<i>Naive benchmarks</i>					
One-step drift	None	11.78 ± 12.81	14.41 ± 14.25	−0.564	0.60 ± 0.43
Constant level	None	17.38 ± 10.13	21.03 ± 10.52	−2.246	1.00

Ridge achieved the lowest average MAE and was the only approach with a positive mean R^2 . The TFT ranked second, followed by the one-step drift benchmark. Neither the FFN nor XGBoost improved on that simple benchmark on average, and the constant-level forecast produced the largest errors by a wide margin. Greater model complexity therefore did not translate into higher forecasting accuracy, even though every estimated model received the same information set, the same 100-day lookback window, the same lag structure, and the same tuning budget.

Answering RQ1. The absolute figures alone do not answer the question of how accurate these forecasts are, because they are not scale-free. The relMAE column supplies that answer: over a 100-day horizon, the best model reduces the mean absolute error to 44% of what a forecaster achieves who simply assumes the deposit stock does not change, averaged across four very different economic environments. The one-step drift benchmark achieves 60%, and the two nonlinear machine-learning models around 70%. In other words, all methods extract genuine signal relative to a do-nothing forecast, but the spread between the best estimated model and a two-line naive rule is far smaller than the spread between either of them and no model at all.

model_comparison.jpeg

Figure 3: Mean level MAE (EUR billions, vertical axis) per model, each in its best-performing covariate specification, averaged over the four rolling origins (2009, 2013, 2019, 2025). Error bars denote one standard deviation across origins. The reference line marks the one-step drift benchmark; only bars below this line improve on a naive rule.

Figure 3 visualises the ranking of Table 4. Only Ridge lies clearly below the naive reference, while the FFN and XGBoost fall on or beyond it. The standard-deviation bars overlap almost completely, which is the visual form of the statistical caveat stated in Section 4.6: with four origins, this ranking describes a sample, not a population.

5.2 Accuracy Versus Robustness

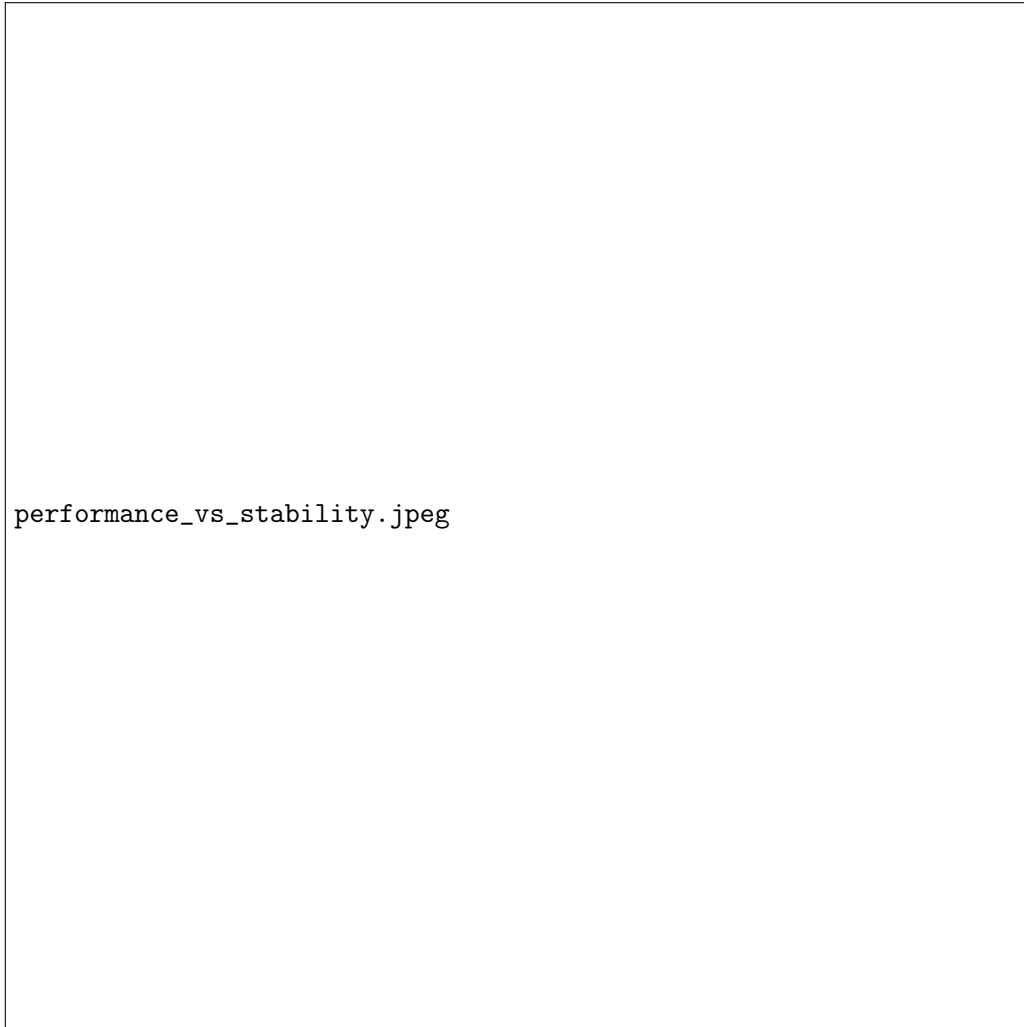


Figure 4: Trade-off between forecast accuracy (mean level MAE, horizontal axis, EUR billions) and regime robustness (standard deviation of level MAE across the four rolling origins, vertical axis). Points closer to the lower-left corner are simultaneously accurate and stable.

Figure 4 shows that, measured in absolute euros, accuracy and robustness are attained by different models: Ridge is the most accurate but has the largest dispersion (± 11.11), whereas the FFN is the most stable (± 6.90) at a distinctly higher error level. No model reaches the lower-left corner.

This apparent trade-off, however, is largely an artefact of the scale problem described in Section 4.6. Once errors are normalised by the constant-level benchmark at the same origin, the ordering changes materially. Ridge attains the lowest dispersion of all estimated models (± 0.31 in relMAE), marginally ahead of the FFN (± 0.34), while XGBoost becomes by far the least stable (± 0.64). Furthermore, the ranking of the two nonlinear models reverses: XGBoost is worse than the FFN in absolute MAE (12.71 versus 11.85) but better in relMAE (0.69 versus 0.71), because its largest absolute errors occur at the origin with the highest deposit level, where they are proportionally less severe. This reversal is a concrete illustration of why a scale-free measure is not a cosmetic addition in a sample whose level roughly doubles.

5.3 Effect of Covariate Selection

The preferred covariate specification differed across models. Ridge performed best with all six explanatory variables (9.36), narrowly ahead of the specification excluding the DAX (9.48). The FFN and TFT both obtained their lowest MAE using only geopolitical risk and inflation (11.85 and 10.78 respectively). XGBoost performed best when inflation was excluded (12.71).

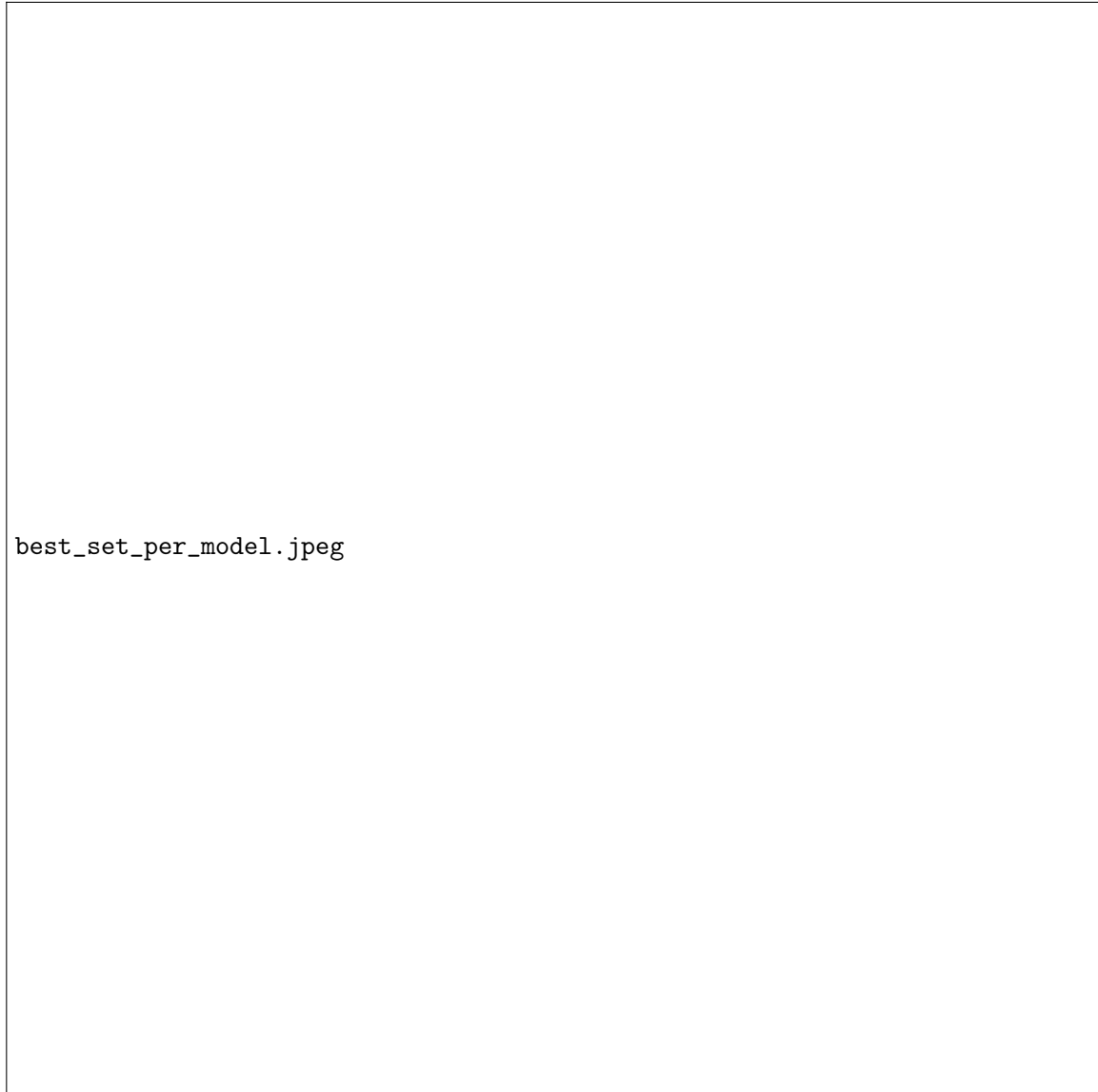


Figure 5: Mean level MAE (EUR billions, vertical axis) of each model across the nine covariate specifications of Table 3, averaged over the four rolling origins. The lowest bar per model identifies the specification reported in Table 4.

Figure 5 shows that sensitivity to the covariate set is itself model-dependent, and dramatically so. Ridge varies within a narrow band across all nine specifications (from 9.36 to 11.81, a spread of 26%), which is the behaviour expected of a shrinkage estimator: adding weakly informative regressors costs little, because their coefficients are pulled towards zero. The FFN varies by a factor of nearly two (from 11.85 to 21.14) and XGBoost by a factor of 1.3 (from 12.71 to 16.05). For the nonlinear models, therefore, the choice of covariate set is a larger determinant of accuracy than the choice of architecture – a result that inverts the usual emphasis of model-comparison studies. The effect is amplified by the input dimensionality that $L = 100$ implies: each additional covariate contributes 100 further columns to the flattened design matrix before lags are counted,

so specification choices translate into large changes in the number of parameters the flexible models must estimate.

We caution against a causal reading. These differences describe how each functional form exploits the available regressors on this sample; they are not evidence that geopolitical risk and inflation *drive* deposit volumes while the deposit rate does not.

5.4 Variation Across Forecast Origins



Figure 6: Level MAE (EUR billions) per covariate specification, resolved by forecast origin (2009, 2013, 2019, 2025). Each panel corresponds to one origin; lower values indicate better forecasts. Note the differing vertical scales across panels.

Forecasting performance varied substantially across the four evaluation periods, and the variation is driven overwhelmingly by a single origin. Averaged over the six model configurations of Table 4, the mean level MAE was approximately 27.9 Bn. € at the January 2009 origin, but only 3.6, 8.3 and 9.4 Bn. € at the 2013, 2019 and 2025 origins respectively. The crisis origin is therefore roughly three times harder than any other and accounts for nearly all of the reported

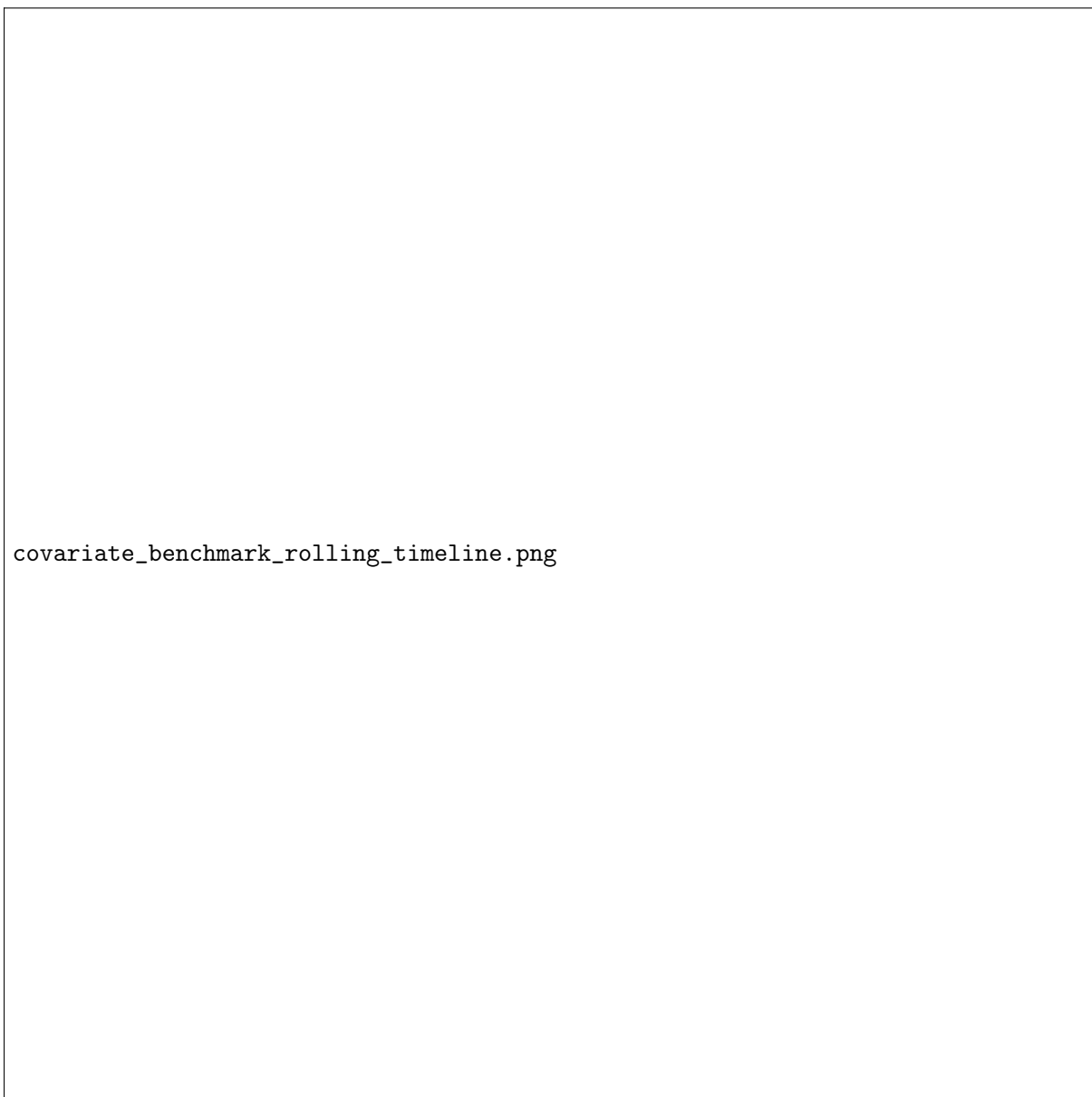


Figure 7: Rolling-origin evaluation: level MAE (EUR billions, vertical axis) per model plotted against the end of the training window (horizontal axis). Naive benchmarks are drawn as dashed reference lines, estimated models as solid lines with markers.



Figure 8: Distribution of level MAE (EUR billions, vertical axis) across the four rolling origins per model, shown as boxplots ordered by median. Box height reflects the interquartile range, whiskers the full observed range across origins.

cross-origin dispersion: excluding it, Ridge’s standard deviation falls from ± 11.11 to well under 3 Bn. €. Figures 7 and 8 make the same point graphically – the error level is dominated by the difficulty of the forecast window, not by the choice of model. It is also the origin with the smallest training sample under the $L + H = 200$ sequence requirement (Section 4.1), so difficulty and data scarcity coincide there.

Two further observations follow from Figure 6. First, no model dominated at every origin. Ridge was best in 2013 and competitive in 2019 and 2025, but the FFN produced the single lowest error at the 2009 crisis origin (20.18 Bn. €, against 25.71 for Ridge and 30.20 for XGBoost), and the TFT the lowest at the 2025 origin (2.43 Bn. €, against 6.05 for Ridge). Second, the covariate specification that minimises MAE also differs by origin for every model, so that the “best sets” of Table 4 are not stable recommendations but sample-specific optima.

5.5 Interpreting the Selection Procedure

Three features of the design mean that Table 4 overstates the achievable accuracy of the estimated models, and we state them here rather than in a footnote.

First, for each estimated model the reported figure is the *minimum* across nine covariate specifications, evaluated on the same four windows used for reporting. A minimum over nine draws is downward-biased even under the null hypothesis that no specification is genuinely better. The naive benchmarks enjoy no such freedom, so the reported margin of Ridge over the one-step drift benchmark (9.36 versus 11.78, or 20.5%) is an upper bound on the margin one should expect out of sample.

Second, the four origins were themselves chosen ex post to represent recognisable regimes, which is defensible as a robustness design but means they are not a random sample of forecast dates.

Third, with $n = 4$ the empirical standard deviations in Table 4 are themselves estimated very imprecisely, and the boxplots of Figure 8 overlap for every pair of estimated models. A formal Diebold–Mariano comparison on four paired observations would have negligible power, and we therefore do not report one.

The appropriate reading of our results is consequently the following: the evidence does *not* establish that Ridge is superior to the TFT, since their intervals overlap. What it does establish, and what is robust to all three caveats above, is the absence of any measurable advantage from the additional complexity, since the more flexible models fail to beat a two-parameter benchmark even when granted the advantage of ex-post specification search that the benchmark is denied.

6 Discussion

The empirical findings from our rolling-origin benchmark provide insight into how distinct algorithmic architectures handle a highly autocorrelated, noisy, and regime-shifting financial series over a 100-day horizon. Contrary to common expectations in the applied machine-learning literature – but consistent with the competition evidence of ? and ? and with the linear-baseline results of ? – a simple regularized linear model was not outperformed by any of the more complex architectures. This section evaluates the structural reasons for that outcome, connects the results to the economic events in the sample, and states the limitations that bound our conclusions.

6.1 Ridge Regression: Accuracy Through Shrinkage

Ridge achieved the highest overall accuracy (MAE = 9.36 Bn. €, relMAE = 0.44), outperforming the one-step drift benchmark by 20.5%, and was the only model with a positive mean R^2 (0.178). Under the scale-free measure it was also the most stable of the estimated models (± 0.31), so the phrase “accurate but volatile” that a reading of the absolute standard deviations would suggest does not survive normalisation.

The structural advantage of ℓ_2 -regularized linear models in cumulative horizon forecasting lies in their smooth penalty mechanism. Because the level forecast is reconstructed by summation (Equation (3)), a small but *systematic* bias in the predicted daily differences accumulates linearly in the horizon: a bias of 0.05 Bn. € per day becomes an error of 5 Bn. € at $h = 100$. What matters over a long horizon is therefore not the variance of the daily prediction but the absence of drift in its mean. The Ridge penalty shrinks weakly identified coefficients towards zero and thereby suppresses precisely the kind of spurious directional signal that would compound.

This mechanism is decisive at the input dimensionality implied by $L = 100$. With a 100-step window, six covariates and three lags each, the flattened design matrix contains on the order of 2,500 columns, estimated from roughly 1,350 overlapping sequences at the earliest origin – a regime in which the ordinary least-squares solution is not merely imprecise but ill-posed. Neighbouring columns are moreover nearly perfectly collinear by construction, since a covariate at lag j and at lag $j + 1$ differ by one day of a smoothed series, and €STR, the deposit rate and the ten-year yield move together over most of the sample. Shrinkage is the textbook remedy for exactly this configuration, and its effectiveness explains why Ridge was simultaneously the most accurate model, the least sensitive to the covariate set (Figure 5), and best served by the full specification, whereas an unpenalised or weakly penalised learner must expend degrees of freedom distinguishing regressors that carry the same information.

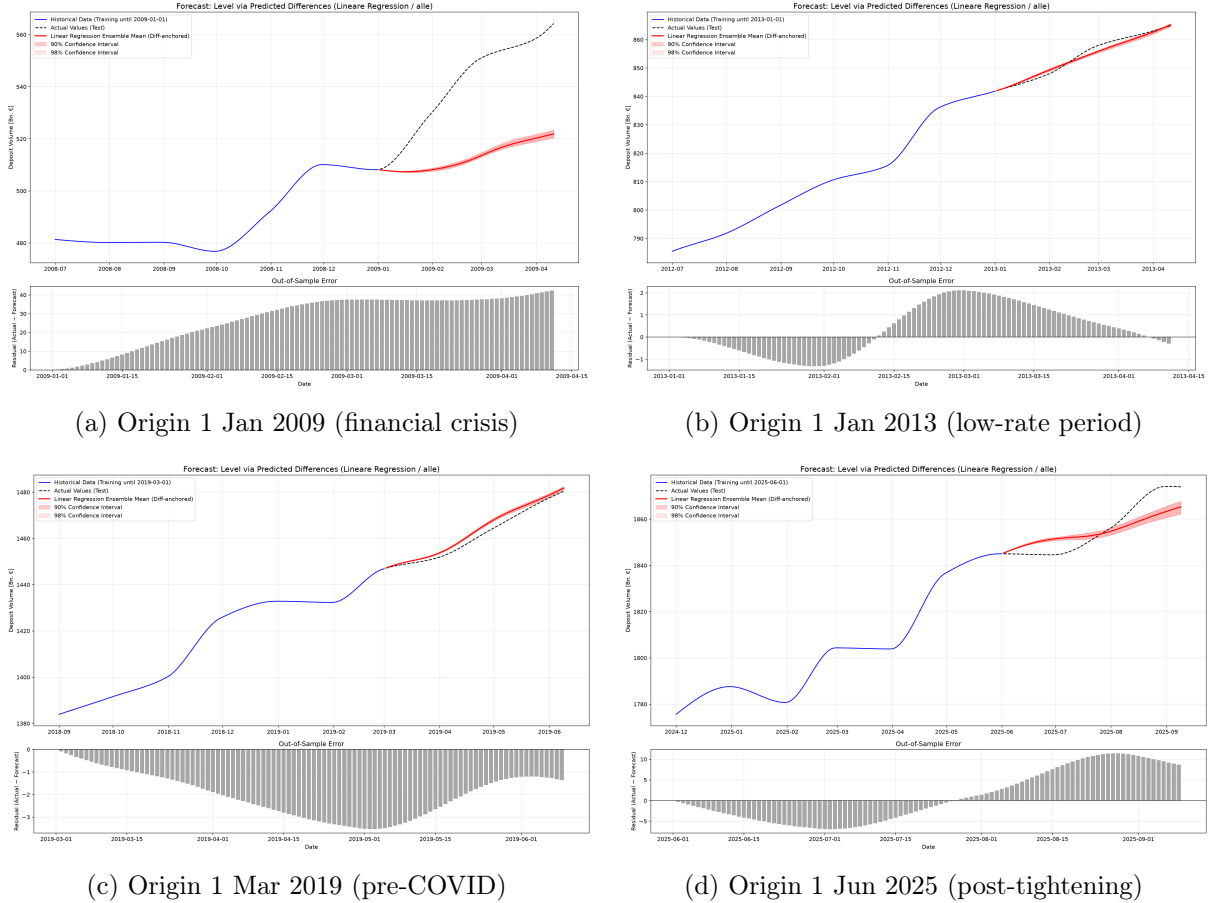


Figure 9: Ridge forecasts with the full covariate set at the four rolling origins. Each upper panel shows the last six months of training data, the realised out-of-sample path, the ensemble mean forecast reconstructed from predicted first differences, and the 90% and 98% ensemble bands; the vertical axis is the deposit volume in EUR billions. Each lower panel plots the daily residual (actual minus forecast) in EUR billions across the 100-day horizon. Note the differing vertical scales across origins.

The residual sub-panels of Figure 9 make this mechanism visible. At the 2013, 2019 and 2025 origins the residuals remain centred near zero and grow only slowly with the horizon, which is the signature of an unbiased difference forecast. At the 2009 origin the residual panel instead shows a monotone, one-sided drift – the model is not noisy there, it is systematically wrong about the direction of the level, and the summation in Equation (3) converts that directional error into the 25.71 Bn. € error reported in Appendix A.

6.2 Temporal Fusion Transformer: Architectural Capacity Without Input Variation

The TFT ranked second overall ($\text{MAE} = 10.78$ Bn. €, $\text{relMAE} = 0.49$) and produced the single most accurate forecast of any model at any origin (2.43 Bn. € in 2025). Although the TFT is designed specifically for multi-horizon forecasting through variable selection and self-attention (?), its potential was structurally constrained by the evaluation setup.

To prevent look-ahead bias, future values of the six macro-financial covariates were held constant at their last observed value over the entire horizon; only the calendar features were supplied as genuine known future inputs. The decoder therefore receives a constant sequence for every economically meaningful variable. The attention mechanism, whose purpose is to weight informative future inputs against one another, has nothing to weight. This is not a defect of the architecture but a property of the forecasting problem: for a bank forecasting at time t , the future path of inflation and market rates is genuinely unknown, and any protocol that supplied it would report accuracy that could not be realised in practice. It does, however, mean that our comparison evaluates the TFT under conditions that neutralise its main design advantage, and the conclusion should be read as “the TFT offers no advantage *when future covariates are unavailable*” rather than as a general statement about the architecture. Notably, the 100-day encoder gives the TFT a genuinely long context over which attention can operate on the historical side, so the constraint lies unambiguously on the decoder input rather than on the amount of history available.

Figure 10 shows the consequence: with frozen exogenous inputs, the predicted path degenerates towards a smooth extrapolation of the encoder trend, and the ensemble bands widen steadily with the horizon as the model signals its own growing uncertainty.

6.3 Feedforward Networks and Gradient-Boosted Trees

Neither the FFN (11.85 Bn. €) nor XGBoost (12.71 Bn. €) consistently outperformed the one-step drift benchmark (11.78 Bn. €), despite both having been tuned under the same protocol and budget as Ridge (Section 4.3). The two models fail in structurally different ways.

- **FFN: consistency without precision.** The FFN exhibited the lowest absolute standard deviation across origins (± 6.90) and produced the best forecast of any model at the difficult 2009 origin (20.18 Bn. €). Its weakness is concentrated in the easy windows: at the 2013 origin, where Ridge achieved 1.00 Bn. € and the TFT 1.05, the FFN required 3.36. In other words, it behaves like a heavily damped forecaster – it never fails catastrophically, but it also never converges tightly on the target when the target is easy. Part of this behaviour is attributable to the input dimensionality: a dense first layer over a flattened 100-step window must estimate a weight for every time step of every covariate, and with a ℓ_2 penalty of at most 10^{-3} that is far weaker than the value Ridge selects, the network has little protection against fitting noise in individual lag positions. For a liquidity manager whose concern is tail behaviour rather than average behaviour, this profile is nonetheless not without value, a point we return to below.
- **XGBoost: the extrapolation trap.** Regression trees partition the feature space into orthogonal hyper-rectangles and predict a constant within each leaf. They therefore cannot

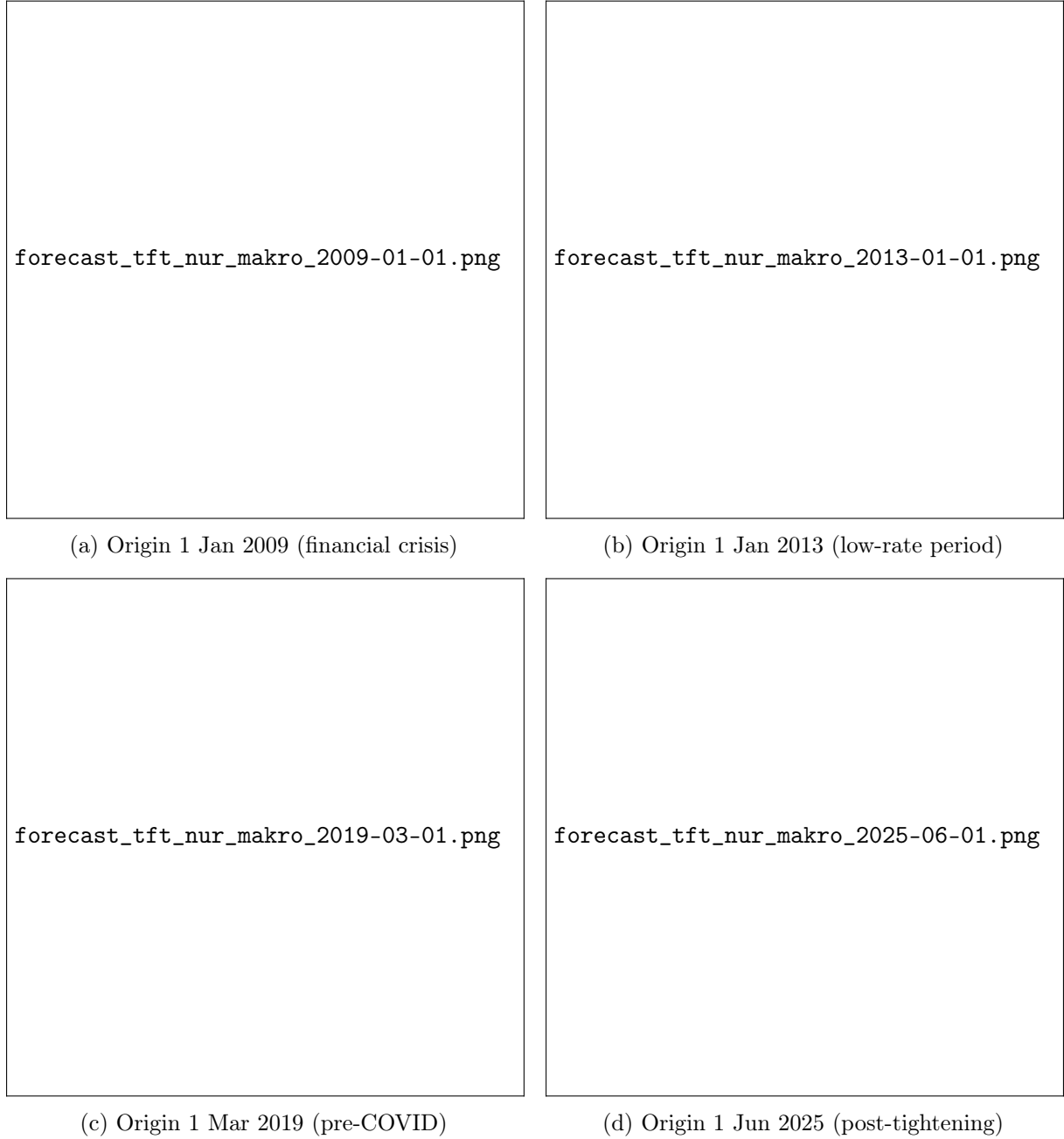


Figure 10: Temporal Fusion Transformer forecasts using the macroeconomic covariate set (geopolitical risk, inflation) at the four rolling origins. Panels are constructed as in Figure 9. Future covariate values are held constant at the last observed level to avoid look-ahead bias, while calendar features enter as genuine known future inputs.

extrapolate a trend beyond the range of differences observed in training. This limitation is most severe exactly where deposit dynamics left their historical range: at the 2025 origin, following the largest reallocation of household deposits since the introduction of the euro, XGBoost recorded 15.77 Bn. € against 6.05 for Ridge and 2.43 for the TFT. XGBoost is also the least stable model under the scale-free measure (± 0.64), which is the same fact seen from another angle.

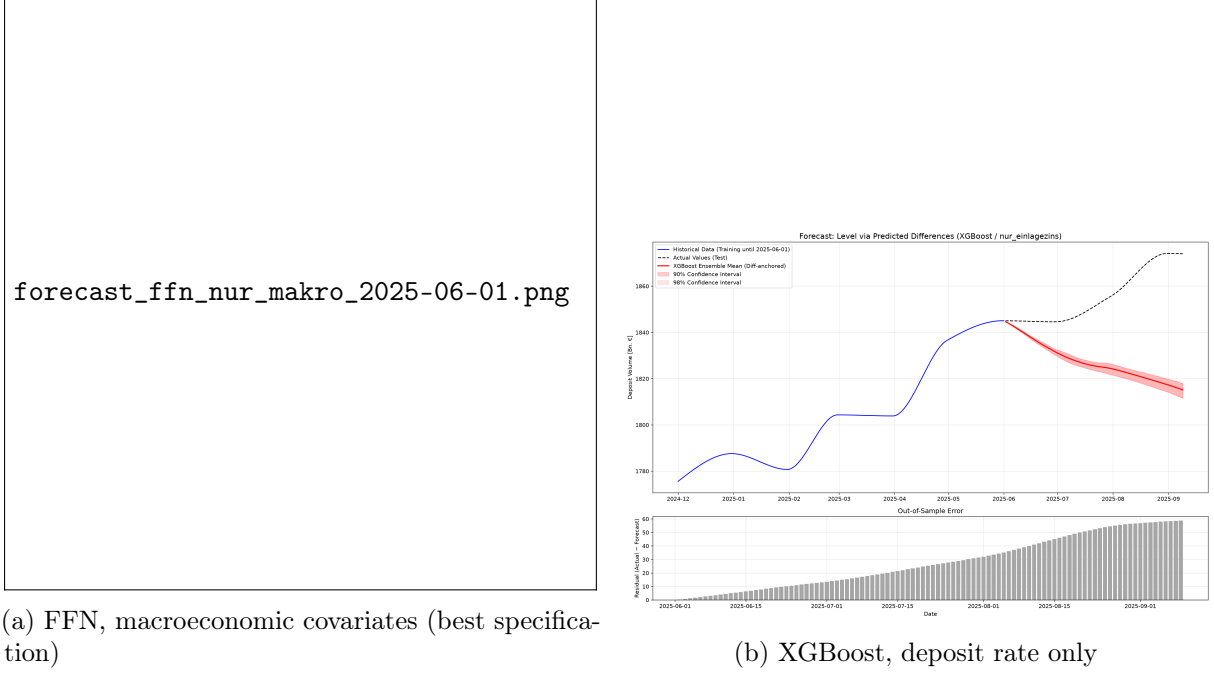


Figure 11: Forecasts of the two nonlinear models at the 1 June 2025 origin, the window most affected by the post-2022 deposit reallocation. Panels are constructed as in Figure 9. Note that the XGBoost panel shows the *deposit-rate-only* specification rather than the best-performing “excluding inflation” specification of Table 4; it is displayed here because it isolates the extrapolation behaviour discussed in the text most clearly.

Figure 11 contrasts the two failure modes directly: the FFN produces a smooth path with a persistent level offset, whereas the XGBoost path flattens after a small number of steps and the residual panel accumulates an almost monotone error across the remainder of the horizon.

6.4 Economic Interpretation: Why 2009 and 2022 Are Hard

The two hardest parts of our sample correspond to the two episodes in which household deposit behaviour was driven by forces that no covariate in our information set measures.

The January 2009 origin sits immediately after the collapse of Lehman Brothers, during a period of acute precautionary saving, deposit guarantees and flight to safety. Deposit volumes moved for reasons of confidence rather than of relative return, and the resulting path is not a function of interest rates, equity prices or measured inflation over the preceding hundred days. Every model, including both naive benchmarks, records its largest error at this origin – the crisis is not a model-selection problem but an identification problem, compounded by the fact that this origin also offers the least training data (Section 4.1).

The second episode shapes the 2025 origin. The 2020–2021 pandemic produced a large accumulation of household overnight balances through forced saving, and the ECB tightening cycle from mid-2022 then triggered the reverse: a substantial reallocation from overnight into term deposits as the spread between them widened for the first time in over a decade. This is

a behavioural response to a *spread* and to a change in the speed of deposit-rate pass-through, neither of which appears as a variable in Table 3 – we include the deposit rate and €STR separately, but not their difference and not the pass-through coefficient between them. That the TFT nonetheless achieved 2.43 Bn. € at this origin while XGBoost required 15.77 reflects the extrapolation argument above rather than superior economic understanding. Constructing an explicit deposit-beta or spread covariate is, in our view, the single most promising extension of this work, and we return to it in Section 7.

6.5 Understanding the Out-of-Sample R^2 Metric

As observed in Table 4, most models yielded negative R^2 values. In multi-step time-series forecasting, R^2 compares predictions against the mean of the test window, $\bar{y}_k$. Because $\bar{y}_k$ is computed from realizations that are unavailable at the forecast origin, it is an unrealistically strong reference: a forecaster who knew the average deposit level over the coming quarter would already possess most of the information being forecast. A negative R^2 therefore indicates that the multi-step trajectory diverged from this post-hoc mean, not that the model has no predictive utility – as the relMAE column confirms, every model beats the genuinely available constant-level reference. Level MAE and relMAE are therefore our primary ranking metrics, and R^2 is reported for comparability with prior work only.

6.6 Limitations

Several features of this study bound the conclusions that can be drawn from it, and we set them out explicitly.

The target is an interpolated series. This is the most consequential limitation. The deposit volume is published monthly and converted to a daily grid by PCHIP interpolation (Section 3.2). Two consequences follow. First, an interpolated value within a month is constructed using the value at the end of that month, which was not known in real time; the constructed “truth” within the evaluation window is therefore partly forward-looking. Second, the resulting daily differences are artificially smooth and strongly autocorrelated, which flatters extrapolative methods – including our one-step drift benchmark – and means that the 100 error terms entering each MAE_k represent roughly three independent observations rather than one hundred. All reported metrics should be read as describing forecast accuracy against a smoothed construct, not against realised daily balances, and a replication at native monthly frequency is the first robustness check we would recommend.

Few origins, and covariate sets selected in-sample. Four forecast origins are too few for inference, and the best covariate set per model is chosen on the same windows on which it is reported. Both issues are analysed in Section 5.5; we report descriptively throughout and avoid claims of statistical superiority.

The tuning budget is equal but not equally generous. Ridge, XGBoost and the FFN were tuned under an identical protocol and an identical budget of at most twelve configurations. The TFT architecture was fixed a priori, because a comparable search was not computationally feasible. A more extensively tuned TFT might close the remaining gap to Ridge. This asymmetry works against our conclusion, but it means that our result is properly stated as “complexity did not pay off *at this level of tuning effort*”.

The lookback window is fixed and not varied. We set $L = H = 100$ for all models on the grounds given in Section 4.1 and did not vary it. Two consequences follow. First, a window of 100 trading days does not span an annual cycle of approximately 250 trading days, so no

model can learn calendar seasonality or interest-rate cycles from its encoder input; monthly and quarterly effects are in any case largely removed by the interpolation of the target. Second, the requirement of $L + H = 200$ observations per training sequence materially reduces the effective sample, and does so most severely at the earliest origin, which is also the hardest. Both effects apply identically to all models and therefore do not bias the comparison between them, but they bound the level of accuracy that is attainable in this design, and it remains possible that the flexible architectures would separate from Ridge given a substantially longer context. A sensitivity analysis over $L \in \{100, 250, 500\}$ would test this directly and is, alongside the monthly-frequency replication, the most important robustness check left open by this study.

Training and inference see different covariate structures. Models are trained on windows in which covariates vary, but forecast with covariates frozen at the origin. This mismatch penalises precisely those models that rely most heavily on covariate dynamics and offers a plausible alternative explanation for why the FFN and TFT achieve their best results with the smallest covariate set. Our RQ3 findings should therefore be read as statements about behaviour under this realistic-information protocol, not as evidence about the intrinsic economic informativeness of the variables.

Aggregate data cannot answer institution-level questions. Our motivation is institution-level ALM, but our data is a sector aggregate published by the Bundesbank. An aggregate series nets out migration between institutions, contains no information on customer segmentation, and reveals neither the deposit beta of an individual bank nor its core/non-core split – which are precisely the quantities that replicating-portfolio and IRRBB frameworks require (???). Our results are therefore best understood as evidence about the predictability of *sector-level* household deposit dynamics, which is useful as a macro-prudential and early-warning input, and as methodological guidance on model selection that plausibly transfers to institution-level data. They are not a validated forecasting system for an individual bank’s deposit book.

Uncertainty bands are not calibrated. The 90% and 98% bands in Figures 9 to 11 are empirical quantiles across ensemble members and therefore quantify estimation uncertainty, not total forecast uncertainty; they are consequently too narrow, and no coverage evaluation is reported. Since liquidity buffer sizing depends on interval accuracy rather than on point accuracy, this is a substantive gap.

Reconstructed €STR history. For observations before March 2017 the €STR series is reconstructed rather than observed, following ?. A substantial portion of the sample therefore contains a synthetic version of one of our central covariates.

6.7 Managerial Implications

For bank asset–liability management and treasury operations, predictability, computational cost, and interpretability are as consequential as raw accuracy. Three implications follow from our results.

First, a regularized linear model is a defensible default for 100-day aggregate deposit forecasting. It was not outperformed by any more complex architecture, its coefficients can be inspected and challenged by a model validation function, it trains in seconds rather than in the minutes to hours required by the TFT, and it is straightforward to document under model-risk governance requirements.

Second, and more importantly, the marginal value of *any* of these models over a naive rule is modest compared with the value of having a model at all. The gap between the best model and the one-step drift rule (relMAE 0.44 against 0.60) is far smaller than the gap between either

and a constant-level assumption (1.00). Institutions currently without a volume model gain most from adopting a simple one; institutions with a simple one should not expect large gains from replacing it with a deep architecture.

Third, model choice should follow the intended use. For central-case planning, the lowest average error is the relevant criterion and favours Ridge. For stress and buffer sizing, what matters is behaviour in the adverse regime, and there the picture differs: at the 2009 crisis origin the FFN was the most accurate model of all, and it exhibited the smallest absolute dispersion across regimes. A treasury function may therefore rationally run a simple model for planning alongside a more heavily damped one for tail-scenario assessment, rather than searching for a single winner.

7 Conclusion and Outlook

Accurate forecasting of non-maturity deposit volumes is fundamental to bank liquidity management, regulatory balance-sheet planning, and interest-rate risk assessment. Modelling aggregate NMDs is nonetheless difficult because the series is non-stationary, strongly autocorrelated, and subject to macroeconomic regime shifts, and because the existing literature treats the volume process largely as a valuation input rather than as a forecasting target evaluated out of sample.

This study evaluated six forecasting approaches across four forecast origins spanning distinct interest-rate regimes (2009–2025) using an expanding-window rolling-origin design, a common information set with a 100-day lookback window and lagged covariates, and a common tuning budget. Our conclusions address the three research questions as follows.

1. **Forecast accuracy (RQ1).** Over a 100-day horizon, the best model reduced the mean absolute error to 44% of the constant-level benchmark ($\text{MAE} = 9.36 \text{ Bn. €}$), averaged across four very different economic environments. Accuracy varied far more across periods than across models: the 2009 crisis origin was roughly three times harder than any other and accounts for nearly all reported cross-origin dispersion.
2. **Model complexity (RQ2).** Greater complexity did not yield higher accuracy. Ridge achieved the best overall performance ($\text{relMAE} = 0.44$, $\text{mean } R^2 = 0.178$) ahead of the TFT (0.49), while XGBoost (0.69) and the FFN (0.71) failed to improve on a one-step drift rule (0.60) despite receiving identical inputs, an identical context length, and an identical tuning budget. Because only four origins are available, we report this as an absence of measurable advantage rather than as demonstrated superiority of the linear model.
3. **Covariate selection (RQ3).** The value of a covariate set depends more on the functional form that consumes it than on its economic content. Ridge varied by only 26% across the nine specifications and was best with the full set, consistent with shrinkage handling the highly collinear design matrix that a 100-step window produces; the FFN varied by a factor of nearly two and was best with the smallest set. Since the optimal set also differs by origin, we treat these as sample-specific optima rather than transferable recommendations.

Outlook and Future Work. Four extensions follow directly from the limitations set out in Section 6.6. First, and most urgently, the benchmark should be replicated at the native monthly frequency of the target, so that accuracy is measured against realised rather than interpolated values. Second, the lookback window should be varied systematically over $L \in \{100, 250, 500\}$: a context spanning at least one full year would allow calendar and cyclical structure to be learned, and would test whether the parity between Ridge and the complex architectures reported here is a property of the task or of the context length we imposed. Third, the frozen-covariate protocol could be replaced by scenario-based path projections for the dynamic covariates, which would restore the TFT’s known-future-input mechanism and permit a fair assessment of its design advantage. Fourth, the economic analysis in Section 6.4 suggests that the informative variable

for the post-2022 period is not the deposit rate or the market rate individually but the spread between overnight and term deposits together with the speed of pass-through; constructing an explicit deposit-beta covariate is, in our assessment, more promising than any further change of architecture. Finally, hybrid designs that combine a regularized linear trend component with a nonlinear residual model may capture sudden systemic liquidity shocks while preserving the stability that made the linear baseline competitive here.

A Appendix: Additional Figures and Tables

Table 5: Complete per-origin results for each model in its best covariate specification. Level MAE in EUR billions; relMAE in parentheses is the ratio to the constant-level benchmark at the same origin. With only four origins, we report the full matrix rather than aggregates alone.

Model	Covariate set	2009-01-01	2013-01-01	2019-03-01	2025-06-01
Ridge	All covariates	25.71 (0.79)	1.00 (0.08)	4.67 (0.33)	6.05 (0.57)
TFT	Macroeconomic	28.44 (0.88)	1.05 (0.09)	11.19 (0.78)	2.43 (0.23)
FFN	Macroeconomic	20.18 (0.62)	3.36 (0.28)	12.77 (0.89)	11.08 (1.04)
XGBoost	Excluding inflation	30.20 (0.93)	1.26 (0.10)	3.63 (0.25)	15.77 (1.49)
One-step drift	None	30.23 (0.93)	2.80 (0.23)	3.43 (0.24)	10.68 (1.01)
Constant level	None	32.40 (1.00)	12.20 (1.00)	14.30 (1.00)	10.62 (1.00)
<i>Mean across models</i>	—	27.86	3.61	8.33	9.44

B Appendix: Code

The complete implementation, including the rolling-origin driver, the tuning protocol of Table 2, and the scripts producing every figure in this paper, is available in the accompanying repository. The excerpt below shows the interface of the shared model-fitting routine.

Listing 1: Shared model-fitting interface.

```

1 def fit_sequence_model(model_name, X_train, y_train, seed=SEED,
2                       bootstrap=False, do_tune=False,
3                       tuned_estimator=None):
4     """Fit one ensemble member of a sequence model.
5
6     model_name : "linreg" | "ffn" | "xgboost"
7     bootstrap : resample training sequences (ensemble members S > 1)
8     do_tune : run the shared grid search (<= TUNING_BUDGET configs)
9     tuned_estimator : reuse a cached configuration for the same origin
10    """
11    ...

```

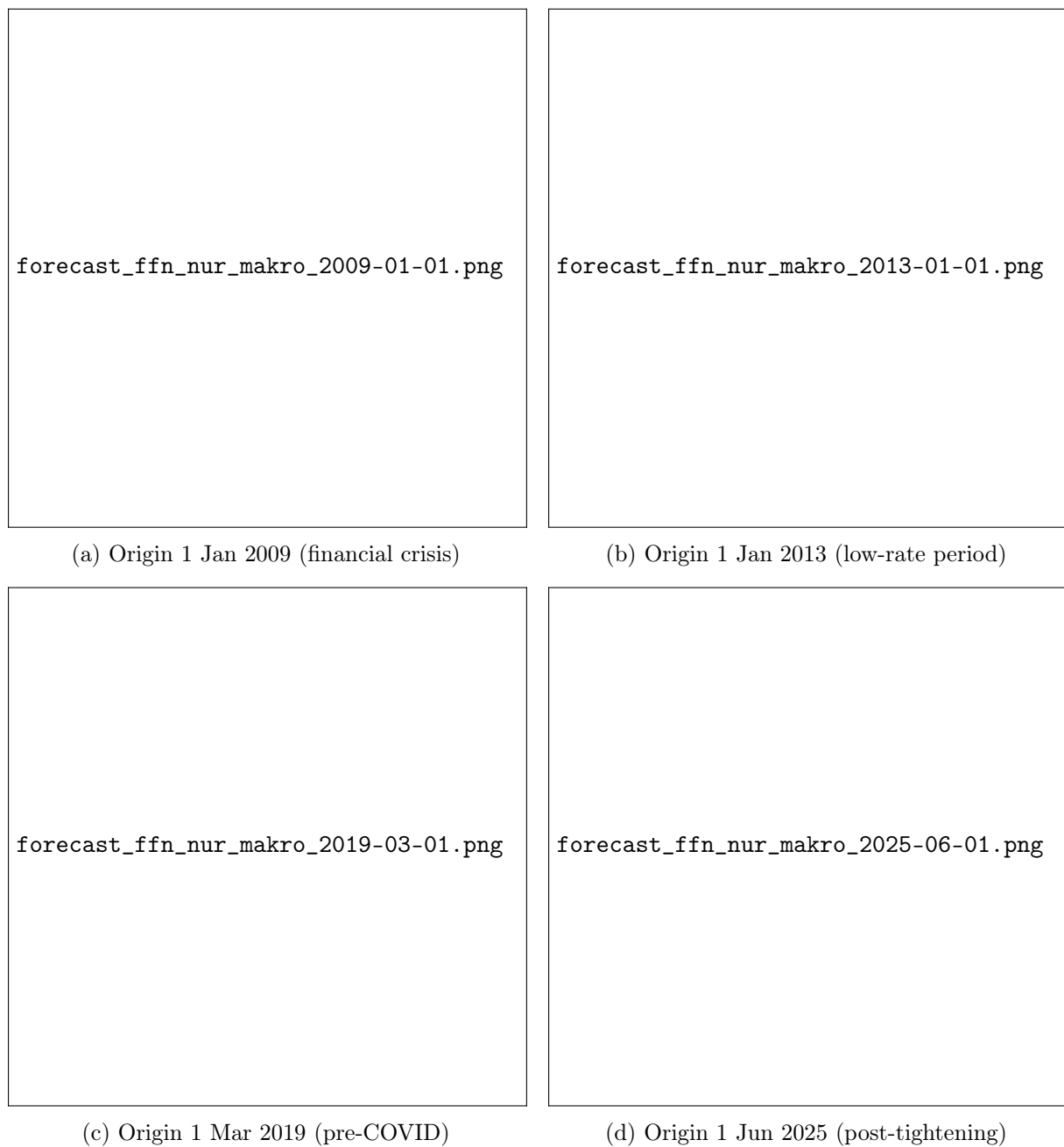
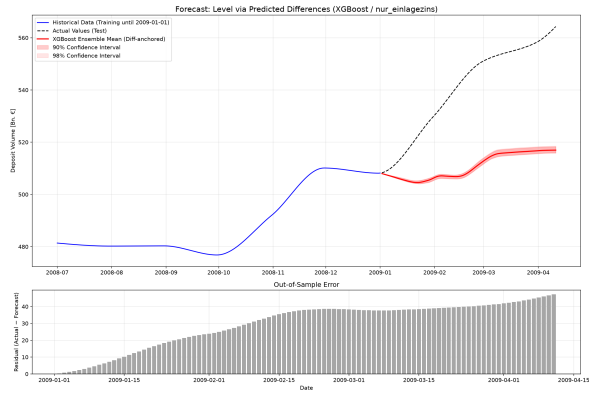
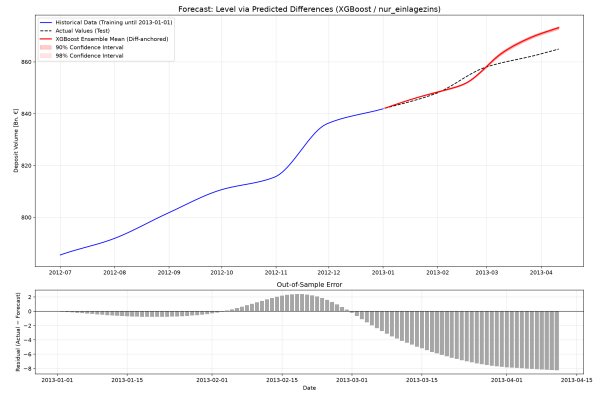


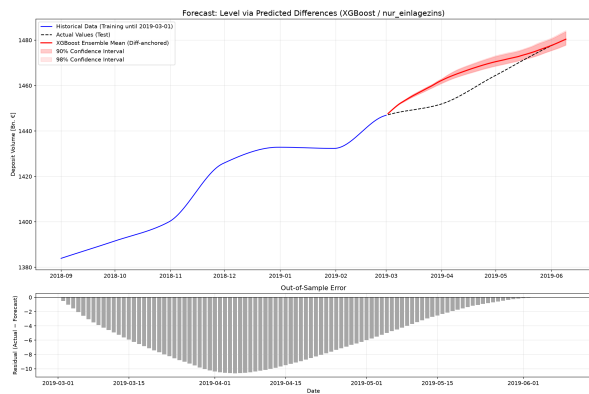
Figure 12: FFN forecasts with the macroeconomic covariate set at all four rolling origins; panel construction as in Figure 9.



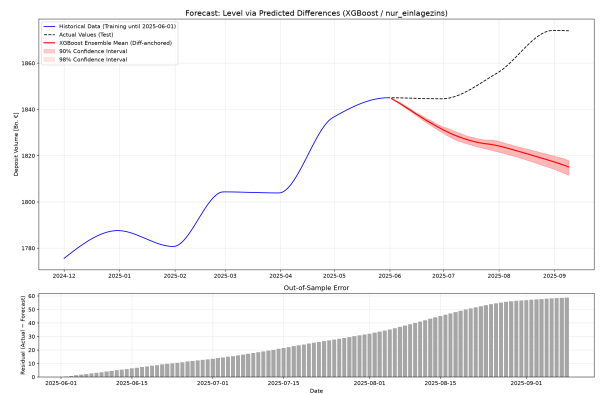
(a) Origin 1 Jan 2009 (financial crisis)



(b) Origin 1 Jan 2013 (low-rate period)



(c) Origin 1 Mar 2019 (pre-COVID)



(d) Origin 1 Jun 2025 (post-tightening)

Figure 13: XGBoost forecasts with the deposit-rate-only specification at all four rolling origins; panel construction as in Figure 9. This is not the best-performing XGBoost specification (see Table 4) but illustrates the extrapolation behaviour discussed in Section 6 most clearly.

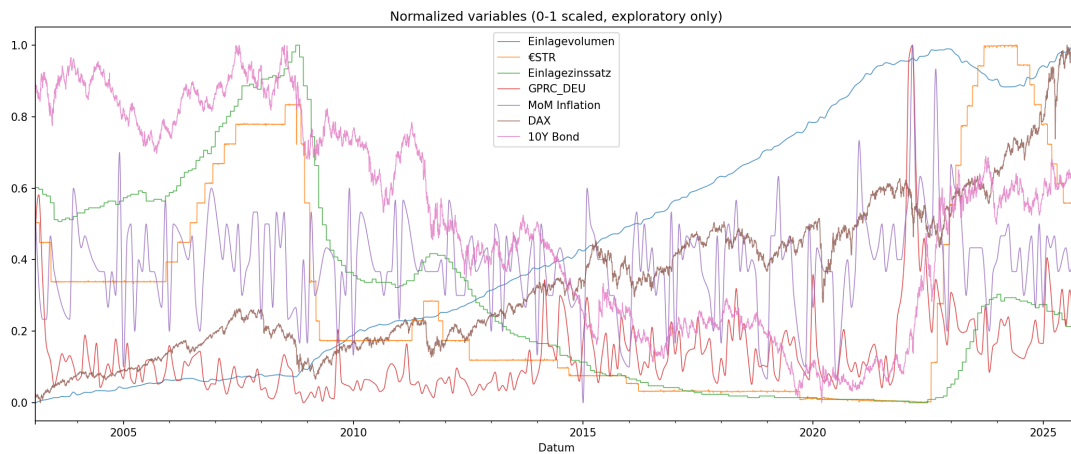


Figure 14: All variables after standardisation to zero mean and unit variance over the sample period January 2003 – September 2025, shown on a common vertical scale so that co-movement between the deposit volume and the covariates is directly visible.